Complete Solution of *Computer Science , Class 9th Federal Board, Morning (Group-I) Paper, 2026*

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Fill the relevant bubble against each question according to curriculum:

	Question	A	B	C	D	A	B	C	D
1.	Which stage of design thinking focuses on understanding users' needs?	Ideating	Empathizing	Prototyping	Testing	○	●	○	○
2.	Why is algorithmic thinking useful while solving a problem?	It provides a step-by-step solution	It removes decision making	It focuses only on outputs	It avoids planning	●	○	○	○
3.	Why should users protect personal data online?	To prevent misuse	To share faster	To reduce cost	To improve graphics	●	○	○	○
4.	Which of the following is on an example of secondary data collection methods?	Experiment	Observations	Interviews	Government record	○	○	○	●
5.	How does a flowchart help in solving a problem?	It stores data	It executes the program	It visually shows the sequence of steps	It replaces algorithms and performs extra calculations	○	○	●	○
6.	Why is identifying inputs, processes, and outputs important in a problem?	To reduce memory usage	To understand how a solution works	To speed up execution	To create diagrams for visualization of the program	○	●	○	○
7.	Why are hyperlinks important in HTML documents?	They connect one page to another	They change page style	They store images	They run scripts	●	○	○	○
8.	Web development is mainly concerned with:	Computer hardware	Managing networks	Installing software	Creating websites and web applications	○	○	○	●
9.	Why are arrays useful in JavaScript?	To store one value only	To store multiple values in a single variable	To change page style and color	To link web pages	○	●	○	○
10.	Why is a computer considered an artificial system?	It grows naturally	It exists in nature	It is man-made to perform tasks	It works without instructions	○	○	●	○
11.	Why is Artificial Intelligence (AI) useful in solving real-world problems?	It stores large files	It improves hardware speed	It replaces the Internet	It enables machines to make intelligent decisions	○	○	○	●
12.	Which issue arises due to the misuse of personal data online?	Increased speed	Improved usability	Privacy violation	Better communica-tion	○	○	●	○

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COMPUTER SCIENCE SSC–I

Inclusive Scheme of Studies 2024 (1200 Marks)

Time allowed: 2:40 Hours **Total Marks Sections B and C: 48**

SECTION – B (Marks 30)

Q. 2 Attempt the following parts. **(10 x 3 =30)**

(i)	Write down the output of the following Javascript code:
	<script type= "text / javascript">
	var a, b, reward;
	reward = 5555;
	a= reward %10;
	b= reward / 10;
	document.write (a, "....", b);
	</script>
		03	OR	Why is packet switching efficient for data transmission? Explain briefly.	03

(ii)	How logical thinking helps in solving a computing problem? Explain briefly.	03	OR	What is cloud computing? Why is it needed in modern computing environment?	2+1
(iii)	Enlist any six Sustainable Development Goals (SDGs).	03	OR	How does machine learning help computers solve problems? Give examples of its applications.	2+1
(iv)	What is meant by output device? Write names of any their output devices.	1.5 + 1.5	OR	What is meant by software piracy? Give one example.	2+1
(v)	What is concept map? How does it represent knowledge?	2+1	OR	Give one use of hyperlinks in HTML documents. Also write down the syntax for defining a link.	1+2
(vi)	How is data transferred in synchronous transmission? Also draw diagram.	2+1	OR	Write down the purpose of any two flowchart symbols.	1.5 + 1.5
(vii)	Briefly describe Volume, Velocity and Variety in Big data.	1x3	OR	How Software engineering differs from Hardware engineering?	03
(viii)	A user receives an email asking for bank details with an urgent warning. Identify the type of cyber threat involved. Justify your answer.	1+2	OR	How is data science helpful in solving business problem?	03
(ix)	Briefly explain the role of front-end development in a web project.	03	OR	Why is the Internet considered a global network of networks?	03
(x)	Briefly discuss primary data collection methods. Also identify any two examples.	2+1	OR	Write HTML code to display unordered list as follows: Software 1. System Software 2. Application Software	03

SECTION – C (Marks 18)

Note: Attempt the following questions. **(3x6=18)**

Q.3	Explain the importance of following in safe and responsible use of computers: a. Backing up Data b. Use of secure Wi-Fi c. Privacy settings	2x3	OR	What is Big data? Write any four benefits of Big data.	2+4
Q.4	How a star topology differs from a bus topology? Explain. Also draw the diagrams.	3+3	OR	How is Artificial Intelligence (AI) used in 'education' and 'gaming' fields?	3+3
Q.5	For an online order system, identify the inputs, processes, and outputs involved.	2+2 +2	OR	Write down the Javascript code to display numbers from 10 to 1.	06

Fbise Notes Publisher(YT and Whatsapp Channel)

SECTION – A (MCQs)

Q.	Correct Option	Logic/Reason
1	**(B) Empathizing**	This stage is about understanding the user's perspective and needs.
2	**(A) It provides a step-by-step solution**	Algorithmic thinking breaks complex problems into manageable steps.
3	**(A) To prevent misuse**	Personal data protection is essential to stop identity theft or fraud.
4	**(D) Government record**	Secondary data is data collected by others (e.g., records, books).
5	**(C) It visually shows the sequence**	Flowcharts use symbols to map out the flow of logic.
6	**(B) To understand how a solution works**	Identifying IPO (Input-Process-Output) defines the logic of the solution.
7	**(A) They connect one page to another**	Hyperlinks are the foundation of navigation in HTML/Web.
8	**(D) Creating websites...**	Web development focuses on building and maintaining web apps.
9	**(B) To store multiple values...**	Arrays allow one variable name to hold a list of data elements.

Q.	Correct Option	Logic/Reason
10	**(C) It is man-made to perform tasks**	Artificial systems are human-engineered, unlike natural systems.
11	**(D) It enables machines...**	AI mimics human intelligence to solve complex real-world problems.
12	**(C) Privacy violation**	Misuse of personal data leads directly to breaches of privacy.

SECTION – B (Short Questions)

Q2 (i)

JavaScript Output:

The code calculates the remainder and the quotient.

- `a = 5555 % 10` → Result: **5**
- `b = 5555 / 10` → Result: **555.5**
- **Output:** `5 ---- 555.5`

OR: Why is packet switching efficient?

Packet switching is efficient because it doesn't require a dedicated circuit. It breaks data into small packets that can take different routes, allowing multiple users to share the same network bandwidth simultaneously.

Q2 (ii)

Logical Thinking in Computing:

It helps identify patterns, break down problems into logical sequences, and predict outcomes. It ensures that the instructions given to a computer are precise and error-free.

OR: Cloud Computing:

Cloud computing is the on-demand delivery of IT resources (servers, storage, databases) over the internet. It is needed for scalability, cost-reduction, and accessing data from anywhere without local hardware.

Q2 (iii)

Six SDGs:

1. No Poverty. 2. Zero Hunger. 3. Good Health and Well-being. 4. Quality Education. 5. Gender Equality. 6. Clean Water and Sanitation.

OR: Machine Learning (ML):

ML helps computers learn from data without being explicitly programmed.

- **Applications:** Face recognition in smartphones, and recommendation systems (like Netflix/YouTube).

Q2 (iv)

Output Device:

A hardware component that conveys information from a computer to one or more people.

- **Examples:** 1. Monitor, 2. Printer.

OR: Software Piracy:

The unauthorized copying, distribution, or use of software.

- **Example:** Installing a paid software using a "cracked" key found online.

Q2 (v)

Concept Map:

A visual tool that represents relationships between concepts. It represents knowledge through nodes (concepts) connected by labeled arrows (relationships) to form a web-like structure.

OR: Hyperlink Use & Syntax:

- **Use:** To navigate from the current web page to another document or section.
- **Syntax:** **<a href="url">Link Text</a>**

Q2 (vi)

Synchronous Transmission:

Data is transferred in a continuous stream at a constant rate. Both sender and receiver are synchronized by a clock signal.

OR: Purpose of Flowchart Symbols:

1. **Oval (Terminal):** Represents the Start or End of a flowchart.
2. **Parallelogram (I/O):** Represents input or output operations.

Q2 (vii)

Big Data (3 Vs):

1. **Volume:** The massive amount of data generated.
2. **Velocity:** The high speed at which data is processed.
3. **Variety:** Different formats of data (text, video, audio).

OR: Software vs. Hardware Engineering:

- **Software Engineering:** Focuses on designing, developing, and maintaining computer programs and systems.
- **Hardware Engineering:** Focuses on the physical components of a computer (circuits, chips, boards).

Q2 (viii)

Cyber Threat (Phishing):

- **Identification:** This is a **Phishing** attack.
- **Justification:** The attacker uses a "urgent warning" and a fake email to trick the user into revealing sensitive bank details.

OR: Data Science in Business:

Data science helps businesses by predicting customer trends, optimizing supply chains, and making data-driven decisions to increase profit.

Q2 (ix)

Role of Front-end Development:

It focuses on the "client-side"—everything the user sees and interacts with (layout, buttons, fonts). It ensures a responsive and user-friendly experience.

OR: Internet as Global Network:

It is a "network of networks" because it connects millions of private, public, academic, and government networks worldwide using the TCP/IP protocol.

Q2 (x)

Primary Data Collection:

Data collected firsthand by the researcher for a specific purpose.

- **Examples:** 1. Personal Interviews, 2. Surveys/Questionnaires.

OR: HTML Code for Unordered List:

```
<ul>
  <li>Software
    <ol>
      <li>System Software</li>
      <li>Application Software</li>
    </ol>
  </li>
</ul>
```

SECTION – C (Long Questions)

Q.3 Safe and Responsible Use:

- **a. Backing up Data:** Protects against data loss due to hardware failure, theft, or malware.
- **b. Use of secure Wi-Fi:** Prevents hackers from intercepting your data (man-in-the-middle attacks) on public networks.
- **c. Privacy Settings:** Allows users to control who sees their personal information, reducing the risk of stalking or identity theft.

OR: Big Data Benefits:

1. **Better Decision Making:** Using analytics to find patterns.
2. **Increased Efficiency:** Identifying bottlenecks in processes.
3. **Personalization:** Tailoring products to specific customer needs.
4. **Fraud Detection:** Identifying unusual patterns in financial transactions.

Q.4 Star vs. Bus Topology:

- **Star Topology:** All devices connect to a central hub/switch. If a cable fails, only one node goes down.
- **Bus Topology:** All devices share a single backbone cable. If the main cable fails, the whole network goes down.

OR: AI in Education & Gaming:

- **Education:** Personalized learning paths (tutors) and automated grading systems.
- **Gaming:** Creating smart NPCs (Non-Player Characters) that adapt to the player's behavior and procedural world generation.

Q.5 Online Order System (IPO):

- **Inputs:** Customer details, Product ID, Quantity, Payment info.
- **Processes:** Validating stock, Calculating total price, Processing payment.
- **Outputs:** Order confirmation, Invoice/Receipt, Shipping label.

OR: JavaScript Code (10 to 1):

JavaScript
```
for (let i = 10; i >= 1; i--) {
  document.write(i + "<br>");
}
```